

Instructions

1. The question paper is divided into two Sections.
2. Please write all the answers of one section together and then write the answers of another section. DO NOT mix the answers from different Sections.

Section A

- ✓ 1. Is it possible for an application to enjoy reliable data transfer even when the application runs over UDP? If so, how? (2 marks)
- ✓ 2. Consider an application that transmits data at a steady rate (for example, the sender generates an N -bit unit of data every k time units, where k is small and fixed). Also, when such an application starts, it will continue running for a relatively long period of time. Answer the following questions, briefly justifying your answer:
 - ✓ a. Would a packet-switched network or a circuit-switched network be more appropriate for this application? Why?
 - ✓ b. Suppose that a packet-switched network is used and the only traffic in this network comes from such applications as described above. Furthermore, assume that the sum of the application data rates is less than the capacities of each and every link. Is some form of congestion control needed? Why or why not? (2+2 marks)
- ✓ 3. Explain slow-start in TCP. What steps are taken by the TCP sender when congestion is detected in the network while it is in the slow-start state? (5 marks)
4. Suppose two packets arrive to two different input ports of a router at exactly the same time. Also suppose there are no other packets anywhere in the router. [NOTE: Justify your answer. Only justification carries marks.] (2x3 marks)
 - ✓ a. Suppose the two packets are to be forwarded to two different output ports. Is it possible to forward the two packets through the switch fabric at the same time when the fabric uses a shared bus?
 - ✓ b. Suppose the two packets are to be forwarded to two different output ports. Is it possible to forward the two packets through the switch fabric at the same time when the fabric uses switching via memory?

- ✓ c. Suppose the two packets are to be forwarded to the same output port. Is it possible to forward the two packets through the switch fabric at the same time when the fabric uses a crossbar?
5. Suppose you are interested in detecting the number of hosts behind a NAT. You observe that the IP layer stamps an identification number sequentially on each IP packet. The identification number of the first IP packet generated by a host is a random number, and the identification numbers of the subsequent IP packets are sequentially assigned. Assume all IP packets generated by hosts behind the NAT are sent to the outside world.
- a. Based on this observation, and assuming you can sniff all packets sent by the NAT to the outside, can you outline a simple technique that detects the number of unique hosts behind a NAT? Justify your answer.
- b. If the identification numbers are not sequentially assigned but randomly assigned, would your technique work? Justify your answer. (3+2 marks)
6. Suppose Bob joins a BitTorrent torrent, but he does not want to upload any data to any other peers (so called free-riding). (2+2 marks)
- a. Bob claims that he can receive a complete copy of the file that is shared by the swarm. Is Bob's claim possible? Why or why not?
- b. Bob further claims that he can further make his "free-riding" more efficient by using a collection of multiple computers (with distinct IP addresses) in the computer lab in his department. How can he do that?

Section B

- ✓ 7. Hosts A and B are each connected to a router R via 10 Mbps links (A ----- R ----- B). The propagation delay on each link is 20 microseconds. R is a store and forward device i.e. it doesn't transmit the packet on the R-B link until the whole packet is received on the A-R link. Suppose R forwards a packet 35 microseconds (processing delay) after it has finished receiving it. Calculate the total time required to transmit 10,000 bits from A to B, in the following two cases: (i) as a single packet, and (ii) as two 5000 bit packets sent one right after the other; show the sequence of events in this case. (2+4 marks)
8. Calculate the total time required to transfer a 1.5 MB file in the following cases, assuming a RTT (Round Trip Time) of 80 ms, a packet size of 1 KB, and an initial 2x RTT of "handshaking" before the data is sent. (2x4 marks)

Note:

- RTT/2 will be the propagation time.
- Assume queueing and processing time to be negligible.

✓ a. The bandwidth is 10 Mbps and data packets are sent continuously.

✓ b. The bandwidth is 10 Mbps, but after we finish sending each data packet we must wait one RTT before sending the next packet.

✓ c. The link allows infinitely fast transmit, that is, transmission time is 0. However, there is a wait time of one RTT after every 20 packets are sent.

✓ d. Zero transmission time as in (c). The first packet is sent and then there is a wait time of one RTT. The next two packets are sent, then there is a wait time of one RTT. The next four packets are sent, then there is a wait time of one RTT, and so on.

✓ 9. Consider sending a 2400-byte datagram into a link that has an MTU of 700 bytes. Suppose the original datagram is stamped with the identification number 422. How many fragments are generated? What are the values of Flag and Offset fields in the IP datagram(s) generated related to fragmentation? (4 marks)

✓ 10. Suppose within your Web browser you click on a link to obtain a Web page. The IP address for the associated URL is not cached in your local host, so a DNS lookup is necessary to obtain the IP address. Suppose that n DNS servers are visited before your host receives the IP address from DNS; the successive visits incur an RTT of $RTT_1, \dots, RTT_n$. Further suppose that the web page associated with the link contains exactly one object, consisting of a small amount of HTML text. Let RTT_0 denote the RTT between the local host and the server containing the object. Assuming zero transmission time of the object, how much time elapses from when the client clicks on the link until the client receives the object? (5 marks)

11. Consider a DASH system for which there are N video versions (at N different rates and qualities) and N audio versions (at N different rates and qualities). Suppose we want to allow the player to choose at any time any of the N video versions and any of the N audio versions. (3x2 marks)

a. If we create files so that the audio is mixed in with the video, so server sends only one media stream at given time, how many files will the server need to store (each a different URL)?

b. If the server instead sends the audio and video streams separately and has the client synchronize the streams, how many files will the server need to store?

✓ 12. Consider a simplified TCP's AIMD algorithm where the congestion window size is measured in number of segments, not in bytes. In additive increase, the congestion window size increases by one segment in each RTT. In multiplicative decrease, the congestion window size decreases by half (if the result is not an integer, round down to the nearest integer). Suppose that two TCP connections, C_1 and C_2 , share a single congested link of speed 30 segments per second. Assume that both C_1 and C_2 are in the congestion avoidance phase. Connection C_1 's RTT is 50 msec and connection C_2 's RTT is 100 msec. Assume that when the data rate in the link exceeds the link's speed, all TCP connections experience data segment loss. (3+2 marks)

a. If both C_1 and C_2 at time t_0 have a congestion window of 10 segments, what are their congestion window sizes after 1000 msec?

b. In the long run, will these two connections get the same share of the bandwidth of the congested link? Explain.